

The metronome and the amplifier: spontaneous cavernosal activity, interstitial cell prostaglandins, and hard flaccid syndrome

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Eighth in a series. Companion papers: *The sacral splanchnic bypass*; *Initiation without maintenance*; *The intraganglionic loop*; *A generator that outlives its cause*; *Autonomy without a signal*; *An amplifier without a signal*; *The wrong co-transmitter*.

Abstract

Background. If the hard flaccid state persists when neural drive is removed, one explanation is that the erectile tissue has acquired its own rhythm. Cavernosal smooth muscle generates spontaneous electrical and contractile activity, and the conjecture examined here proposes that this intrinsic oscillator has been chronically upregulated by sustained noradrenergic exposure.

The conjecture misidentifies the oscillator. It attributes pacemaking to interstitial cells within the corpus cavernosum. The source it cites states the opposite: interstitial cells in the genitourinary system are not a simple analogy of the interstitial cells of Cajal that act as electrical pacemakers in the gut, and in bladder and corporal tissue — where the smooth muscle cells are themselves capable of generating spontaneous electrical activity — interstitial cells *modulate* smooth muscle activity. More recent work localises the oscillator explicitly: TMEM16A calcium-activated chloride channels underlie spontaneous electrical activity in isolated corpus cavernosum smooth muscle cells. **The metronome is the myocyte.**

The interstitial cells do something else, and it is more useful. In corporal tissue, interstitial cells release prostaglandins through cyclooxygenase-2 activity, reinforcing both spontaneous excitation of cavernosal smooth muscle and nerve-mediated α -adrenergic contractions, while endothelial nitric oxide suppresses excitation. Cavernosal tone is set by the balance between the two. This reframes the conjecture from an untestable claim about pacemaker cells into a claim about a prostaglandin–nitric oxide balance — with the notable practical consequence that cyclooxygenase-2 is the only target named anywhere in this series for which licensed, widely used, orally available human drugs already exist.

The evidential base is thinner than it looks. The finding that noradrenaline accelerates pacemaker activity roughly twelve-fold (1.5 to 18 depolarisations per minute) is from rabbit *urethral* interstitial cells, not cavernosal tissue. No study has examined spontaneous cavernosal activity after nerve injury or chronic adrenergic exposure, in any species.

Tests. The conjecture's distinctive prediction — that resting tone should be oscillatory rather than steady — has an existing instrument. Spontaneous corpus cavernosum electromyography has been recorded in humans and applied clinically, though it remains formally experimental and is contaminated by cutaneous sympathetic potentials. We propose it as the primary test, with an organ-bath study of spontaneous activity after cavernous nerve injury as the necessary animal counterpart.

Conclusion. Reformulated around myogenic pacemaking and interstitial-cell prostaglandin amplification, this conjecture moves from the most speculative in the series to one of the few with both a plausible instrument and an available drug class.

Keywords: hard flaccid syndrome; spontaneous activity; interstitial cells; TMEM16A; cyclooxygenase-2; prostaglandins; corpus cavernosum electromyography.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. In particular, nothing in this paper supports taking non-steroidal anti-inflammatory drugs or cyclooxygenase-2 inhibitors for this condition; these carry cardiovascular, gastrointestinal and renal risks, and the reasoning here is a research proposal, not a treatment rationale. Nothing here should influence clinical management.

1. Introduction

Four papers in this series have looked for the source of a contraction that reportedly persists after sympathetic blockade [5, 6, 7]. Each has proposed that something about the effector has changed: its calcium sensitivity, its responsiveness,

or its transmitter balance. This paper examines the most radical member of the family — that the tissue has stopped needing an input at all, because it has acquired its own rhythm.

The premise is sound and often overlooked. Cavernosal smooth muscle is not a passive tissue awaiting instructions. Smooth muscles in the lower urinary tract and corporal tissue exhibit spontaneous contractile activity dependent on L-type calcium channels, and the mechanism underlying it is spontaneous electrical activity [C] [8]. Penile detumescence is maintained by tonic contraction of cavernosal smooth muscle cells [C] [9]. If that intrinsic activity were chronically upregulated, the tissue would be contracted without a signal — and removing the signal, as a nerve block does, would change nothing.

The conjecture proposes that this is what has happened, and that the responsible structures are the interstitial pacemaker cells of the corpus cavernosum [2]. That attribution is incorrect, and correcting it turns out to improve the hypothesis rather than damage it.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *interstitial cells corpus cavernosum*, *spontaneous activity cavernosal smooth muscle*, *TMEM16A corpus cavernosum*, *interstitial cells of Cajal lower urinary tract pacemaker*, *corpus cavernosum electromyography spontaneous*, and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic. We read the source the conjecture cites in order to check what it says, and Section 3 reports the result.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence, or a physiological principle established across independent laboratories and preparations.
[B]	Single human study, human tissue study, or human data with substantial methodological limitation.
[C]	Experimental animal or in vitro evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. The correction: the metronome is the myocyte

The conjecture states that interstitial pacemaker cells within the corpus cavernosum generate the spontaneous rhythmic activity that sets baseline tone, and that these cells become chronically upregulated [2]. Its cited source says something materially different.

Interstitial cells are widely distributed in the genitourinary system and are implicated in spontaneous muscle activity, but they **are not a simple analogy of the interstitial cells of Cajal in the gut, which act as electrical pacemakers**; they represent variability among tissues. In the bladder and corporal tissue, **where smooth muscle cells are themselves capable of generating spontaneous electrical activity, interstitial cells may modulate smooth muscle activity** [C] [8].

More recent work localises the oscillator to a specific molecular mechanism in the myocyte. In freshly isolated mouse corpus cavernosum smooth muscle cells, TMEM16A calcium-activated chloride channels underlie spontaneous electrical activity [C] [9]. The same channel is the downstream effector of purinergic input in this tissue: P2X1 receptor activation on cavernosal myocytes produces calcium influx that activates TMEM16A [C] [VERIFY] [14], a convergence noted in the companion paper on co-transmission [7].

The distinction is not pedantic. It determines what would have to change for the conjecture to be true, what tissue would have to be examined, and what could be targeted. **Under the conjecture as written, one would look for proliferation or upregulation of a distinct cell population. Under the corrected account, one would look for altered myogenic excitability — and separately, for altered modulation of it.**

3.1 What the interstitial cells actually do here

The corrected role is more specific and, for present purposes, more useful. In corpus cavernosum, both spontaneous activity of cavernosal smooth muscle and neurally released noradrenaline contract the muscle. Spontaneously produced prostaglandins, generated via cyclooxygenase-2 (COX-2) activity in interstitial cells, *reinforce both* the spontaneous excitation of cavernosal smooth muscle *and* nerve-mediated α -adrenergic contractions. Conversely, spontaneously released nitric oxide from the endothelium suppresses excitation. The balance between spontaneously released prostaglandins and nitric oxide regulates cavernosal smooth muscle tone and determines the contractile state of the penis [C] [8].

This is not an isolated claim. Endothelium-derived nitric oxide and cyclooxygenase products were shown to modulate corpus cavernosum smooth muscle tone more than three decades ago [C] [VERIFY] [13].

Three consequences follow [D]:

1. **The interstitial cell contribution is an amplifier, not a clock.** Upregulated COX-2 activity would raise tone by reinforcing an oscillator it does not generate.
2. **It amplifies both arms.** Prostaglandins reinforce spontaneous excitation *and* α -adrenergic contraction. A hypothesis built on this predicts that neither pure sympathetic blockade nor pure α -blockade would normalise tone, because the amplification survives both.
3. **The target is druggable now.** COX-2 is inhibited by widely used, licensed, orally available drugs. This is the only mechanism proposed anywhere in this series for which that is true, and it is the reason we think the corrected version of this conjecture deserves more attention than the original.

We immediately qualify the third point. That a target is druggable is a statement about feasibility of *investigation*, not about therapy; we make no treatment claim, and Section 7.3 sets out why enthusiasm here would be premature and potentially harmful.

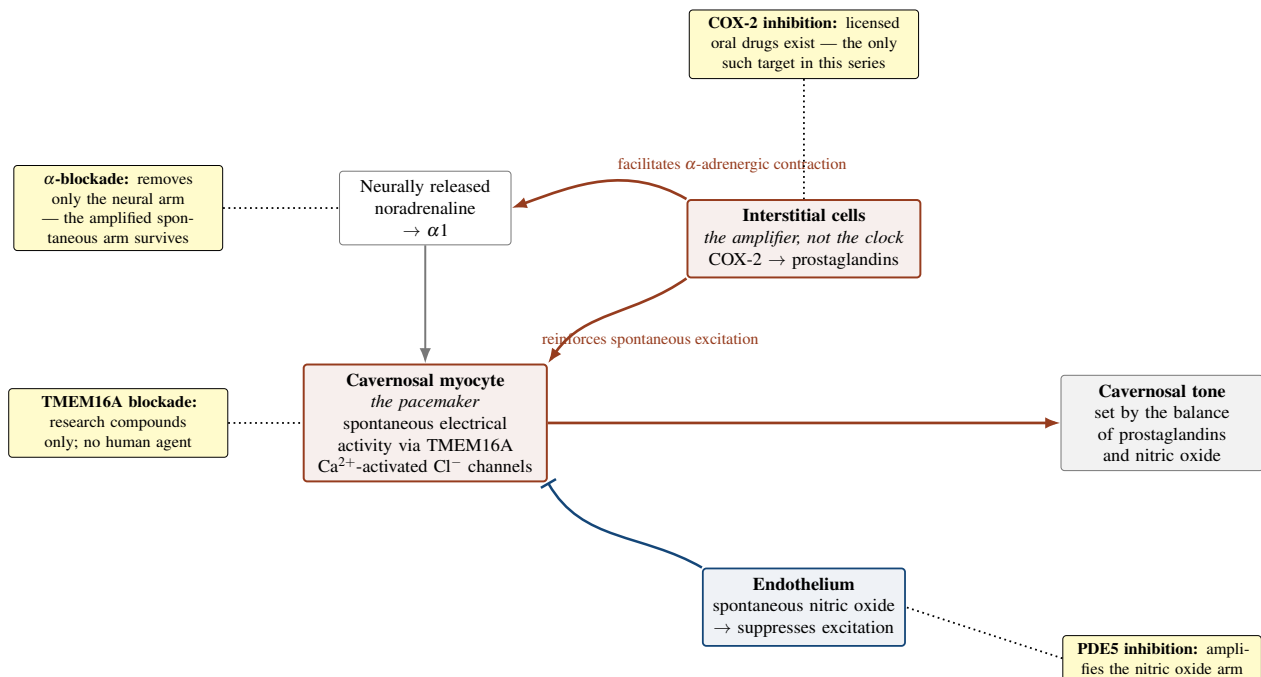


Figure 1. Regulation of cavernosal tone as described in the source the conjecture cites, redrawn. The oscillator is the cavernosal myocyte itself, whose spontaneous electrical activity depends on TMEM16A calcium-activated chloride channels [9]. Interstitial cells do not pace it; they amplify it, releasing prostaglandins through cyclooxygenase-2 activity that reinforce both the spontaneous excitation of cavernosal smooth muscle and nerve-mediated α -adrenergic contraction, while endothelial nitric oxide suppresses excitation [8]. Note that α -blockade removes only one of the two arms the prostaglandin signal amplifies, which is the shape of result reported clinically. Diagram is schematic; the underlying data are from animal preparations.

4. The hypothesis, reformulated

Formal statement

H8 (spontaneous activity dysregulation). In hard flaccid syndrome the intrinsic spontaneous activity of cavernosal smooth muscle is elevated, and/or its amplification by interstitial-cell-derived prostaglandins is increased relative to endothelial nitric oxide suppression. The tissue therefore generates and sustains contraction without neural input.

H8-myogenic. The oscillator itself is upregulated — increased frequency or amplitude of spontaneous electrical activity in cavernosal myocytes, plausibly through altered TMEM16A function or calcium handling. Prediction: spontaneous cavernosal electrical activity is increased in frequency or amplitude; resting tone fluctuates on the oscillator's timescale rather than being steady.

H8-amplifier (the form we recommend). The oscillator is normal but its prostaglandin amplification is increased, or nitric oxide suppression reduced. Prediction: tone is reduced by cyclooxygenase inhibition; the amplification survives both sympathetic blockade and α -blockade because it reinforces the spontaneous arm as well as the neural one.

H8-cellular (the form as originally proposed). Distinct interstitial pacemaker cells are upregulated and drive the tissue. Contradicted by the account in Section 3, in which corporal interstitial cells modulate rather than pace.

H8-null. Spontaneous cavernosal activity in HFS is normal. Whatever maintains the state is neural, structural, or not present.

Auxiliary assumptions:

- **A1 (induction).** Sustained noradrenergic exposure or nerve injury upregulates spontaneous cavernosal activity or its prostaglandin amplification. **Never examined in cavernosal tissue in any species** [DATA NEEDED]. This is the hypothesis's first step and it is unexamined rather than merely unproven.
- **A2 (magnitude and persistence).** The change is large enough, and lasts long enough, to produce a symptomatic state over years. Untested.
- **A3 (human relevance).** Human cavernosal tissue behaves like the animal preparations in which this physiology was characterised. The interstitial-cell and TMEM16A data are from guinea-pig, rabbit and mouse [8, 9]; the human evidence is limited to recordings of spontaneous cavernosal electrical activity whose interpretation is contested (Section 6).

5. Where the conjecture's key detail actually comes from

The conjecture makes much of a specific finding: that the pacemaker is inhibited by nitric oxide through cyclic GMP and *enhanced by noradrenaline* [2]. It is a genuinely striking result and the arithmetic is worth stating. In current-clamp recordings, interstitial cells fired spontaneous depolarisations at a frequency of about 1.5 per minute, rising to 18 per minute in the presence of 10 μ M noradrenaline — a roughly twelve-fold acceleration; the depolarisations were abolished in calcium-free solution [C] [VERIFY] [11].

These are rabbit urethral interstitial cells, not cavernosal tissue. Related work in rabbit urethral smooth muscle showed spontaneous activity dependent on calcium release from intracellular stores, with spontaneous calcium transients accelerated by noradrenaline and suppressed by a nitric oxide donor [C] [10]. In the urethra, interstitial cells of Cajal do act as putative pacemakers, generating spontaneous global calcium waves [C] [VERIFY] [12].

So the conjecture has assembled a mechanism from the urethra and applied it to the corpus cavernosum, in a field where the same author has explicitly warned that interstitial cells across the genitourinary system are *not* a single functional class [8]. The urethral finding remains suggestive — noradrenaline accelerating an intrinsic oscillator twelve-fold is exactly the kind of mechanism the conjecture needs — but it is evidence about a different tissue with a different pacemaking arrangement [D].

6. The instrument the conjecture needs already exists

The conjecture's distinctive prediction is that the hard flaccid state should not be perfectly steady but should fluctuate on a timescale set by the oscillator rather than by neural drive [2]. Alone among the fourteen conjectures, this one predicts a *temporal signature* rather than a magnitude — and unlike the tone measurements required elsewhere in this series, an instrument for it exists.

Spontaneous electrical activity of the human corpus cavernosum has been recorded with needle electrodes during flaccidity and erection since the late 1980s [B] [VERIFY] [15], and its nature has been the subject of dedicated study [B] [VERIFY] [16]. Spontaneous corpus cavernosum electromyography has been applied to assess cavernosal smooth muscle status in erectile dysfunction, with reported potential amplitudes in the region of 200–250 μ V in some patients [B] [VERIFY] [17].

The limitations are serious and must be stated in the design, not discovered afterwards. A consensus review of neurophysiological assessment in male sexual dysfunction concluded that corpus cavernosum electromyography remains experimental and that abnormalities can be attributed both to damage to autonomic penile innervation and to degenerative processes in cavernous smooth muscle [B] [VERIFY] [18]. Penile surface potentials are contaminated by cavernous smooth muscle activity and vice versa, and one study concluded that surface potentials from penile skin should not be used to evaluate cavernosal autonomic innervation [B] [19]. Lack of standardisation of recording technique and of objective criteria for characterising the signals are the principal obstacles to clinical application [B] [VERIFY] [18].

None of this precludes using the method as a research instrument for a within-study comparison against matched controls, provided the analysis is of the temporal structure of the signal — frequency, regularity, burst pattern — rather than of an absolute amplitude threshold, and provided contamination controls are recorded simultaneously. The signal-processing question the conjecture asks is precisely the kind that a contested amplitude measure can still answer, because oscillatory structure survives calibration uncertainty in a way that absolute magnitude does not [D].

7. Predictions and tests

7.1 Primary human test: temporal structure of spontaneous cavernosal activity

Prediction P1. Compared with age-matched controls, men with HFS will show increased frequency, increased regularity, or altered burst structure of spontaneous cavernosal electrical activity, recorded over a prolonged period at rest.

Design requirements: prolonged recording rather than a snapshot, since the prediction concerns rhythm; simultaneous recording of a cutaneous sympathetic channel to control for the contamination documented by Soyulu and colleagues [19]; blinded analysis; pre-specified spectral or interval-based outcome measures; and a matched control group, since no normative dataset exists [DATA NEEDED]. Test–retest reliability within subjects should be reported, as it has not been established for this application.

Prediction P2. Any objective measure of resting penile tone, once one exists [3, 5], will show oscillatory rather than flat elevation, with the oscillation period matching that of the electrical activity.

7.2 Necessary animal test: does spontaneous activity change after injury?

Prediction P3. In cavernosal tissue from animals subjected to cavernous nerve injury or chronic adrenergic exposure, spontaneous electrical and contractile activity is increased in frequency or amplitude relative to sham controls.

This is a conventional organ-bath and patch-clamp study in preparations that have already been used to characterise this physiology [9, 8]. **Nothing else in this paper is worth doing until it has been done**, because A1 — that anything about spontaneous cavernosal activity changes after injury — is currently unexamined rather than merely unproven.

7.3 The prostaglandin test

Prediction P4. Cyclooxygenase inhibition reduces spontaneous cavernosal activity and resting tone, and does so in tissue in which α -blockade has failed.

In principle this is the most accessible pharmacological probe proposed anywhere in this series, because the drug class is licensed, oral and familiar. In practice we recommend that it be pursued *in vitro and in animals first*, for four reasons.

1. The prostaglandin contribution has been characterised in animal corporal tissue [8, 13]; its magnitude in human corpus cavernosum has not been quantified [DATA NEEDED].
2. Prostaglandins in erectile tissue are not uniformly contractile. Prostaglandin E1 is a standard intracavernosal *vasodilator*. Which prostanoid species and which receptors dominate under the conditions of interest is unresolved, and non-selective cyclooxygenase inhibition would suppress relaxant as well as contractile prostanoids, with an uncertain net effect [D].
3. Non-steroidal anti-inflammatory drugs and coxibs carry cardiovascular, gastrointestinal and renal risks that are not trivial in long-term use, and this is a young population with a non-life-threatening condition.
4. **There is a specific risk of harm from publishing this reasoning.** Anti-inflammatory medication has already been suggested for penile symptoms in this condition on quite different grounds [B] [1], and a mechanistic-sounding rationale is exactly the sort of thing that drives self-medication in an underserved patient community. We therefore state plainly: *this is a hypothesis about a target, not evidence that any drug helps, and it should not be acted on.*

7.4 What would discriminate this hypothesis

Prediction P5. The oscillatory signature is unique to H8. No other conjecture in this series predicts that resting tone should fluctuate with a characteristic period; the neural hypotheses predict tone tracking neural drive, and the

calcium-sensitization and supersensitivity hypotheses predict a steady elevation [4, 5, 6]. **A demonstration of flat, non-oscillatory elevated tone would refute H8 while leaving its siblings intact** — which makes this among the cleanest discriminations available in the series.

Table 2. Predictions of H8 compared with the principal alternatives.

Observation	H8 (spontaneous activity)	Calcium sensitization	Neural drive (any route)	Discriminating value
Resting tone oscillates with a characteristic period	Predicted	Not predicted	Not predicted	High
Flat, non-oscillatory elevated tone	Refutes	Predicted	Predicted	High
Increased frequency of spontaneous cavernosal electrical activity	Predicted	Not predicted	Not predicted	High
Confirmed sympathetic block fails to reduce tone	Predicted	Predicted	Not predicted	Low
Cyclooxygenase inhibition reduces tone	Predicted (amplifier form)	Not predicted	Not predicted	Moderate–high
Spontaneous activity increased after nerve injury in animals	Predicted	Not predicted	Not predicted	High, and unexamined
Response to PDE5 inhibitors	Predicted (NO arm suppresses the oscillator)	Predicted	Not specified	None

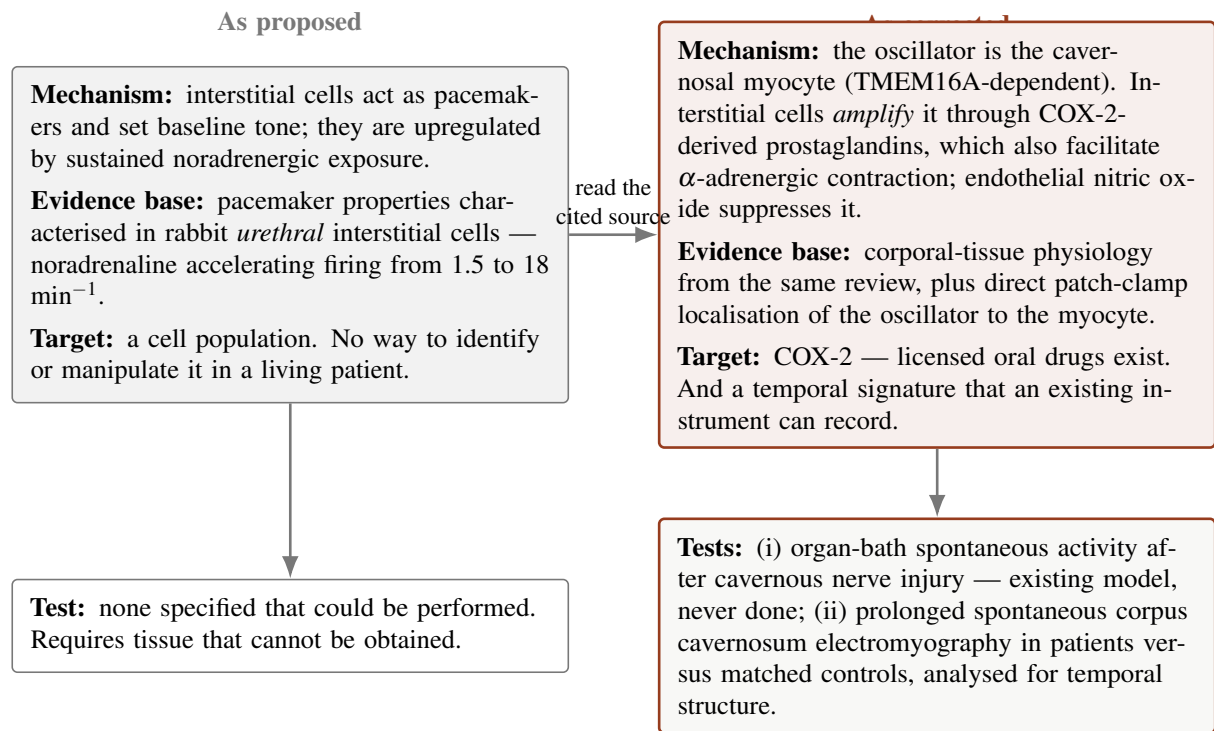


Figure 2. The conjecture as proposed and as corrected. Correcting the attribution of pacemaking from interstitial cells to the cavernosal myocyte does not weaken the hypothesis; it relocates the interstitial cell contribution to a prostaglandin pathway that is both better supported in corporal tissue and, uniquely in this series, pharmacologically accessible. It also leaves the conjecture’s distinctive prediction — an oscillatory rather than steady tone — intact, with an instrument that already exists.

8. Alternative explanations

- Technical failure of the blocks.** As in every effector hypothesis in this series, the motivating observation is grade E and undocumented [2, 3]. If sympathetic blockade was never adequate, there is no autonomy to explain.
- Reported variability may be ordinary.** Patients describe symptom fluctuation that does not track obvious triggers, and the conjecture treats this as weak evidence of the right kind [2]. Symptom variability in a distressing, closely self-monitored condition has many explanations — attentional, postural, diurnal, affective — none requiring an oscillator. **Self-reported fluctuation is not evidence of a myogenic rhythm, and should not be cited as though it were.**

3. **Calcium sensitization.** Would produce steady rather than oscillatory elevation [5], and is distinguished from H8 by exactly that.
4. **The oscillator may be normal and the brake broken.** Reduced endothelial nitric oxide would raise tone by removing suppression of an entirely normal oscillator [8, 2]. This is a distinct hypothesis that H8's tests would not separate from it, since both predict reduced tone with PDE5 inhibition and neither predicts response to α -blockade.
5. **The premise may be false.** Cavernosal tone has never been objectively shown to be elevated in HFS, and its temporal structure has never been examined in anyone.

9. Implications

No clinical implications are drawn, and we repeat that nothing here supports taking anti-inflammatory drugs for this condition. Three research implications follow.

First, read the source before building on it. The conjecture's central attribution is contradicted by the review it cites. This is worth stating as a general point about hypothesis generation in an underserved field: mechanisms assembled from abstracts and secondary summaries accumulate errors that look like insight.

Second, do the animal experiment. Whether spontaneous cavernosal activity changes after nerve injury has apparently never been asked, and can be answered in preparations already used for exactly this physiology.

Third, record the rhythm, not the amplitude. Corpus cavernosum electromyography is a contested amplitude measure but a usable temporal one. The question this conjecture asks — is the tone oscillatory? — is answerable with an instrument whose calibration problems do not matter for that question.

10. Limitations

1. **No evidence in HFS.** Spontaneous cavernosal activity has never been measured in a patient with this condition.
2. **The first step is unexamined.** No study in any species has asked whether nerve injury or chronic adrenergic exposure alters spontaneous cavernosal activity.
3. **The conjecture as stated is contradicted,** not merely unsupported, by its own cited source on the question of what interstitial cells do in corporal tissue.
4. **Its key quantitative finding is from the wrong tissue.** The twelve-fold noradrenergic acceleration is a urethral interstitial cell result.
5. **The species base is animal throughout.** Guinea-pig, rabbit and mouse preparations; the human evidence is limited to a contested electrophysiological recording method.
6. **The instrument is formally experimental,** unstandardised, without normative data for this application, and contaminated by cutaneous sympathetic potentials.
7. **The prostaglandin arm may not be contractile in humans.** Prostaglandin E1 is a clinical intracavernosal vasodilator; which prostanoids dominate in this tissue under resting conditions is unresolved, and cyclooxygenase inhibition could act in either direction.
8. **Patient-reported fluctuation is not evidence for the hypothesis,** and the conjecture's use of it should be discounted.
9. **Several load-bearing citations are flagged unverified,** having been traced through secondary sources or figure legends rather than read in full.
10. **Confirmation risk.** This paper was written to develop a specific conjecture, and it concludes by recommending a reformulation that makes the hypothesis more attractive — including by identifying an available drug class. That is exactly the direction in which motivated reasoning runs, and readers should weigh Section 7.3 accordingly.

11. Refutation criteria

- **Refuted.** Prolonged objective measurement shows resting cavernosal tone in HFS to be flat and non-oscillatory.
- **Substantially refuted.** Spontaneous cavernosal electrical activity in men with HFS is indistinguishable in frequency and temporal structure from matched controls.
- **Substantially refuted.** Organ-bath studies show no change in spontaneous cavernosal activity after cavernous nerve injury or chronic adrenergic exposure.

- **Weakened (amplifier form).** Cyclooxygenase inhibition has no effect on spontaneous or resting cavernosal tone in animal corporal tissue.
- **Already contradicted (cellular form).** Corporal interstitial cells modulate rather than pace cavernosal smooth muscle.
- **Rendered moot.** Direct measurement shows resting cavernosal tone is not objectively elevated in symptomatic patients.

12. Conclusion

The source article calls this the conjecture with the widest gap between mechanistic elegance and evidential support, and rates it lowest of the effector hypotheses. We think that assessment is right about the evidence and wrong about the elegance, because the mechanism as stated is not the one the literature describes.

Corporal interstitial cells are not pacemakers. The review the conjecture cites says so explicitly, distinguishing them from the interstitial cells of Cajal that pace the gut, and locating spontaneous electrical activity in the cavernosal smooth muscle cells themselves — a localisation since made molecular, with TMEM16A calcium-activated chloride channels underlying spontaneous activity in isolated cavernosal myocytes. The metronome is the myocyte. What the interstitial cells do is amplify: they release prostaglandins through cyclooxygenase-2 that reinforce both the spontaneous excitation and the nerve-mediated α -adrenergic contraction, against an endothelial nitric oxide signal that suppresses both. Cavernosal tone is the balance of the two.

Corrected, the hypothesis is better rather than worse. It no longer requires a cell population nobody can identify in a living patient. It explains why neither sympathetic blockade nor α -blockade would normalise tone, since the prostaglandin amplification reinforces the spontaneous arm as well as the neural one. It keeps its distinctive and genuinely discriminating prediction — that the tone should have a rhythm — and that prediction has an instrument, because spontaneous cavernosal electromyography exists and its calibration problems do not compromise a question about temporal structure. And it names the only target in this entire series for which licensed oral human drugs already exist.

That last point is the one we are most anxious about. An accessible drug class attached to a mechanism-shaped story is precisely what causes harm in a community of young men with a distressing condition and few options. The prostaglandin arm has not been quantified in human tissue, prostanoids in this organ are not uniformly contractile, and the entire first step of the hypothesis — that anything about spontaneous cavernosal activity changes after injury — has never been tested in any species. That experiment is straightforward, uses preparations that already exist, and should come first.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** Because this paper argues that the conjecture misreads its own principal source, that source should be read in full and independently confirmed before submission.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [2]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. **Nothing in it supports the use of non-steroidal anti-inflammatory drugs, cyclooxygenase-2 inhibitors or any other medication for hard flaccid syndrome.** These drugs carry cardiovascular, gastrointestinal and renal risks and should be taken only as directed by a treating clinician for an established indication. Corpus cavernosum electromyography is an invasive research procedure requiring ethical approval.

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